

DASC7606C Group Project

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Overview

The group project accounts for **25%** of your final grade in DASC7606C. Based on initial enrollment, the class will be divided into **10 groups** of approximately **10–12 students**. Group sizes may be adjusted if enrollment changes.

We recommend forming your group after the first class. Once you have agreed on your group members and a tentative topic, nominate one representative to register through the Google Form [DASC7606C Group Project Registration](#). Submit one registration per group. If your group is not yet full, list your current members and indicate how many additional members you need. The teaching team will help coordinate group allocation. Or you can also join others' groups you are interested in this way.

After the teaching team confirms your group registration, submit one proposal per group covering:

- Your chosen project track, specific topic, and objectives.
- Each member's role and responsibilities.
- Your planned approach and expected deliverables.

Before Lecture 5.

All group members must review and agree to the proposal before submission. The proposal will be reviewed for scope and feasibility but will not receive a separate grade. The TAs will contact your group if revisions are needed.

Your topic is fixed once the proposal is approved. Any subsequent topic change requires approval from the teaching team. You may refine your methods and deliverables as the project develops, provided they remain consistent with the approved topic.

Please follow the registration and proposal **deadlines** announced on **Moodle**.

If you have any questions, please contact:

- All Staff: dasc7606c-staff-2026fall@googlegroups.com
- Head TA: teaching@bangjunwang.com

GPU Resources

- Option 1: You can apply for access to the HKU CS GPU Farm (GPU Farm for Teaching, formerly Phase 2) - RTX 4080 Cluster. Please refer to <https://www.cs.hku.hk/research/major-facilities?view=article&id=130&catid=24> for resource applications and usage.
- Option 2: DASC7606C is supported by [Moore Threads](#) by offering our students with higher standard state-produced GPU resources. **If you can integrate Moore Threads GPUs into your group project and include the profiling and benchmarking results in your final report, your group will get a 5-point bonus directly on the final project group score.**

More details regarding Option 2's bonus points will be announced later.

If you have any questions regarding GPU resources, please contact Serena for help via serena.cds@hku.hk

Policy on LLM Usage

You may use large language models (LLMs) for project ideation, writing, coding, and analysis. In fact, we encourage you to embrace AI tools while using them thoughtfully and checking their outputs carefully.

Adapted from the [ICLR 2027](#) policy, the following rules apply:

1. Disclose all LLM use. Include an “LLM Usage Statement” in your proposal and final report. Identify the tools used, the specific sections or components affected, and how the tools contributed. This includes grammar and wording improvements. If no LLMs were used, state this.
2. Take responsibility for your work. Your group is responsible for all submitted content and must verify LLM-generated claims, references, code, and results. Using an LLM does not excuse false claims, plagiarism, fabricated data, or misrepresented results.
3. Protect confidentiality. Do not upload confidential information or restricted data to an LLM without authorization.
4. Do not manipulate assessment. Hidden prompts or instructions intended to influence grading are prohibited, whether or not they succeed.

Disclosed LLM use will not itself reduce your grade. Violations will be handled under the course and institution’s academic integrity policies.

We will conduct strict checks for AI-related plagiarism and undisclosed AI use. Suspected violations will be reviewed using supporting evidence, rather than AI-detection scores alone. Confirmed misconduct may result in substantial penalties to your final course grade, including a zero for the group project, and further disciplinary action under the University’s academic integrity policies.

Example LLM Usage Statement

- Section 2 — Market Analysis: We used ChatGPT to identify potential competitors. We checked the product descriptions and pricing included in the report against official sources.
- Section 3 — Technical Approach: We used Claude to compare candidate model architectures. The group selected the final approach and verified the cited references.
- Prototype — preprocess.py: We used ChatGPT to draft the data-cleaning function. We modified the function and tested its handling of missing values and duplicate records.
- Sections 1 and 5 — Writing: We used ChatGPT to improve grammar and clarity. The group reviewed and edited the suggested wording.

Your statement should describe your group’s actual use; the examples above are illustrative.

Topics

This group project offers four tracks. The number of 🌟 indicates each track's difficulty and potential for higher marks. Choosing a more difficult track does not guarantee a higher score: an excellent submission in a simpler track can outperform one that only meets the basic requirements of a harder track. Your grade will depend on the quality of your work and how well it meets the assessment criteria.

Track 1: AI Startup Business Proposal 🌟

Propose an AI startup whose core product uses deep learning. Explain the customer problem, how the technology solves it, and how the startup could earn revenue.

Requirements

Your proposal must cover:

1. Problem and customers: Who needs the product? What problem does it solve? Provide evidence that the problem matters.
2. Product: Describe the product, how customers would use it, and its advantages over existing alternatives.
3. Deep learning approach: Explain the model, inputs and outputs, data sources, and evaluation metrics. Justify your technical choices.
4. Business model: Explain pricing, customer acquisition, and estimated development and operating costs. State your assumptions.
5. Feasibility and risks: Outline an MVP, development milestones, and major technical, business, or ethical risks.

Pretrained models and commercial APIs are allowed. A working prototype or demo is encouraged but not required. However, the proposal must include evidence of technical feasibility, such as results from a small experiment or relevant published research.

Deliverables

- Report: 5–8 pages, excluding references.
- Pitch: 10-minute presentation followed by 5-minute Q&A.
- Contribution statement: A brief description of each member's work.
- LLM disclosure.

Scoring

Criterion	Points	What we assess
Problem and customer need	20	Clear target customers and evidence of a meaningful problem
Product and competitive advantage	15	Useful product and convincing comparison with alternatives
Deep learning approach	25	Appropriate models, credible data plan, and sound evaluation
Business model	20	Plausible pricing, customer acquisition, and costs
Feasibility and risks	10	Realistic scope, supporting evidence, and awareness of risks
Report, presentation, and Q&A	10	Clear explanation and well-supported answers
Total	100	

Cite external claims and clearly label assumptions. Marks reward evidence and sound reasoning; ambitious claims and polished slides alone will not earn high scores.

Track 2: Agentic Framework Design 🌟🌟

Design and build an AI tool that solves a specific problem. Explain who it serves, how it works, and demonstrate its usefulness through a working demo.

Requirements

Your project must cover:

1. Problem and users: Who will use the tool? What problem does it solve? Explain why the problem matters.
2. Tool design: Describe the main features, user workflow, and advantages over existing alternatives.

3. Deep learning approach: Explain the models, inputs and outputs, data sources, and evaluation metrics. Justify your technical choices.
4. Implementation and evaluation: Build a working tool and evaluate it using representative examples or test cases. Report what works and where it fails.
5. Limitations and risks: Discuss technical limitations, privacy concerns, and other relevant risks.

Pretrained models and commercial APIs are allowed. Your contribution should go beyond a basic model or API call by providing a usable workflow for the chosen problem.

A working demo is required and must be shown during the presentation. Demonstrate the complete workflow, from user input to the resulting output. Prepare a backup recording in case of technical problems.

Deliverables

- Report: 5–8 pages, excluding references.
- Presentation: 10 minutes, including the demo, followed by 5 minutes of Q&A.
- Code: Source code with setup instructions and any necessary sample inputs. Do not include API keys or restricted data.
- Contribution statement: A brief description of each member's work.

Scoring

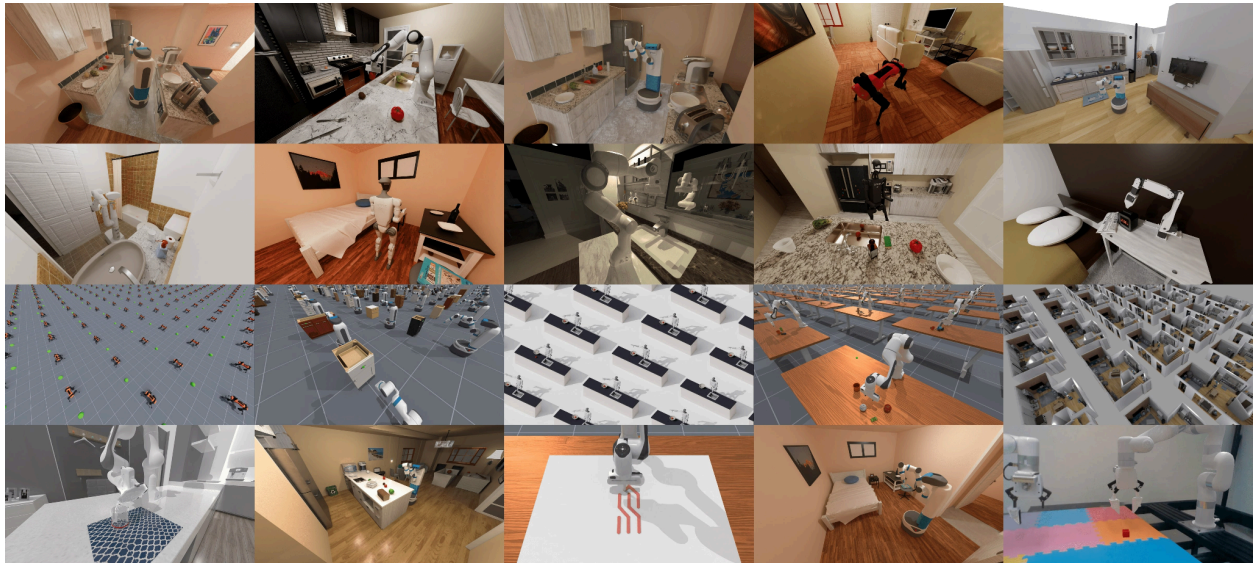
Criterion	Points	What we assess
Problem and user need	15	Clear users and a meaningful problem
Tool design and usability	15	Useful features and a coherent user workflow
Deep learning approach	20	Appropriate models, justified choices, and a credible data plan
Implementation and demo	25	A functioning tool that demonstrates the intended workflow
Evaluation and limitations	15	Meaningful testing, evidence of usefulness, and honest analysis of failures and risks
Report, presentation, and Q&A	10	Clear explanation and well-supported answers

Total	100	
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Cite external sources and follow the course's LLM usage policy. Marks reward a useful, reliable tool and sound technical reasoning. A complex system does not automatically earn a higher score.

Track 3: Simulation Experiments 🌟🌟

Collect or generate expert demonstrations and train and evaluate **Diffusion Policies (DP)** on at **least six distinct ManiSkill tasks**. Investigate a research question beyond basic training and demonstrate your findings.



Resources

- ManiSkill: [Official repository](#) · [Documentation](#)
- Diffusion Policy: [Original repository](#) · [Project website](#) · [Paper](#)

Use the documentation matching your installed ManiSkill version. The original DP repository is a reference implementation; integration with ManiSkill may require adaptation.

Requirements

1. Task selection: Select at least six distinct tasks covering different manipulation skills. Different random seeds of the same task do not count as separate tasks. Describe the observations, action spaces, and success criteria.
2. Expert demonstrations: Collect or generate demonstrations for all six tasks using teleoperation, motion planning, scripted controllers, or another justified method. Document the process, dataset sizes, and quality checks. Existing tools are allowed, but downloading datasets alone does not satisfy this requirement.
3. Policy training: Train and evaluate a DP baseline for every task. Separate task-specific policies are acceptable. Document the architecture, preprocessing, training settings, and computing resources.
4. Further investigation: Explore at least one question from the suggestions below, or propose your own. This investigation may focus on a justified subset of tasks.
5. Evaluation and analysis: Use held-out evaluation seeds or initial conditions. Report per-task success rates, evaluation episode counts, and failure cases. Keep comparisons fair and state any differences in data or compute budgets.

Suggested Research Questions

- Co-training: Does training one policy on multiple tasks help or hurt performance compared with separate policies? How do task similarity, data balance, and task conditioning affect the results?
- Improving DP: Can changes to the observation encoder, prediction horizon, action representation, or diffusion sampling improve success rates or inference speed? Test individual changes against the baseline.
- Cross-embodiment: Can demonstrations from one robot help train a policy for another? Explore how to handle different observations, action spaces, and robot geometries. Scope this study to compatible tasks.
- Data efficiency: How many demonstrations are needed? Does demonstration diversity or quality matter more than quantity?
- Generalization: How well does the policy handle changes in object positions, appearances, or physical properties? Can augmentation or domain randomization improve robustness?

You are not expected to explore every direction. A focused investigation with clear evidence is sufficient, even if the proposed change does not improve performance.

Deliverables

- **Report:** 5–8 pages, excluding references.
- **Presentation:** 10 minutes, including a **required demo**, followed by 5 minutes of Q&A. Show the demonstration-generation process and policy rollouts, with video coverage of all six tasks. Recordings are acceptable.

- **Code, data, and policies:** Collection scripts, generated datasets or download links, training and evaluation code, configurations, and trained checkpoints, with reproduction instructions.
- **Contribution statement:** A brief description of each member's work.

Scoring

Criterion	Points	What we assess
Task coverage and demonstration generation	20	Six distinct tasks, a working collection process, and documented data quality
DP implementation and baseline training	20	Correct implementation and documented training across all six tasks
Further investigation	20	A clear question, justified approach, and controlled experiments
Evaluation and analysis	25	Fair comparisons, per-task results, and thoughtful interpretation of failures
Demo and reproducibility	5	Clear demonstrations and usable artifacts
Report, presentation, and Q&A	10	Clear explanation and well-supported answers
Total	100	

Follow the course's LLM usage policy and acknowledge external code and data. Marks reward experimental rigor and understanding; high success rates alone do not guarantee a high score.

Track 4: Real-robot Training & Deployment 🌟🌟🌟

Collect real-world demonstrations on a Unitree G1 humanoid robot through teleoperation using a Noitom motion-capture suit. Train a robot policy using HKU computing resources, deploy the trained policy on the physical robot, and evaluate it on real-world manipulation tasks. NVIDIA Isaac GR00T N1.7 is recommended.

Resources

- Isaac GR00T N1.7: [Official repository](#) · [Fine-tuning guide](#) · [Deployment guide](#) · [Model checkpoint](#)

- GEAR-SONIC: [Official repository](#) · [Documentation](#) · [Collect–train–deploy workflow](#) · [VLA inference](#)

Use the documentation matching the software versions installed on the course machines. The teaching team may provide additional starter code for Noitom teleoperation, motion retargeting, and robot control.

Requirements

1. Select a distinct real-world manipulation task with clearly defined initial conditions and success criteria. At least one task should involve bimanual coordination, multiple stages, or meaningful variation in object position or appearance. Tasks must be approved based on feasibility and safety.
2. Use the Noitom motion-capture suit to teleoperate the Unitree G1 and collect original demonstrations for all selected tasks. Describe the calibration and motion-retargeting process, observations, actions, control frequency, reset procedure, dataset size, and quality checks. Existing datasets may supplement but may not replace your own data collection.
3. Convert the demonstrations into the format required by the selected model. For GR00T N1.7, the recommended pipeline uses LeRobot-format data. Document synchronization, preprocessing, normalization, filtering, language annotations, and train-validation splits. Report how many trajectories were collected, rejected, and retained.
4. Fine-tune GR00T N1.7 using HKU computing resources. Document the pretrained checkpoint, model inputs and outputs, training configuration, computing resources, checkpoint selection, and training results.
5. Deploy the trained policy on the Unitree G1 and demonstrate closed-loop inference. Explain the deployment architecture, communication between the inference computer and robot, control frequency, inference latency, action limits, and emergency stopping mechanism. SONIC may be used as the whole-body controller.
6. Explore at least one research question from the suggestions below, or propose your own. The investigation may focus on a subset of the tasks but must include a controlled comparison.
7. Evaluate the policy using initial conditions not included in the training demonstrations. Report task success rates, evaluation episode counts, inference latency, completion time where relevant, and representative failure cases. Keep comparisons fair and include videos of both successful and failed trials.
8. All physical experiments must follow the laboratory's safety procedures and be conducted under authorized supervision. Use an emergency stop, conservative motion limits, and a designated safety operator. Validate predicted actions offline and test at reduced speed before autonomous execution. Unapproved locomotion or unsafe human–robot contact is prohibited.

Suggested Research Questions

- Data efficiency: How many demonstrations are needed? Does demonstration diversity or quality matter more than quantity?
- How many real-world demonstrations are needed, and how do diversity and quality affect performance?
- How does GR00T N1.7 compare with another policy architecture under a similar data and compute budget?
- How do joint-space, end-effector, or SONIC latent actions affect training and deployment?
- How well does the policy handle changes in object position, appearance, lighting, or task instruction?
- Can action chunking, temporal ensembling, inference acceleration, or remote inference reduce latency without reducing success?

You are not expected to explore every direction. A focused investigation with clear evidence is sufficient, even if the proposed method does not improve performance.

Deliverables

- **Report:** 5–8 pages, excluding references.
- **Presentation:** 10 minutes, including a required demo, followed by 5 minutes of Q&A. Show the teleoperation process and autonomous policy rollouts for every selected task. Recordings are acceptable and preferred when a live demonstration would create unnecessary risk.
- **Code, data, and policies:** Data-collection and processing code, training and evaluation code, configurations, deployment scripts, collected datasets or approved access instructions, trained checkpoints, and reproduction instructions.
- **Videos:** Representative teleoperation demonstrations and successful and failed autonomous trials for every task.
- **Contribution statement:** A brief description of each member's work.

Scoring

Criterion	Points	What we assess
Teleoperation and data collection	20	A working collection process, original demonstrations, correct calibration, and documented data quality
Policy training	20	Correct implementation, justified model choices, and documented training
Real-robot deployment	20	Reliable closed-loop inference, system integration, and measured deployment performance

Further investigation	15	A clear question, controlled experiments, and evidence-supported conclusions
Evaluation and failure analysis	15	Fair testing, per-task results, generalization, and thoughtful analysis of failures
Safety, demo, and reproducibility	5	Safe operation, clear demonstrations, and usable artifacts
Report, presentation, and Q&A	5	Clear explanation and well-supported answers
Total	100	

Follow the course's LLM usage policy and acknowledge all external code, models, and datasets. Marks reward a complete and safe real-robot learning pipeline, experimental rigor, and technical understanding; a polished demonstration or high success rate alone does not guarantee a high score.

Further Clarification about Grading

The group project will be assessed by the instructor and TA panel, who will assign an overall score to each group. Each member's individual score will be calculated by multiplying the group score by a contribution factor, based on their contribution statement and supporting evidence. **Scores will not be announced publicly.** Once officially communicated, individual scores will be considered final, subject to the University's applicable review procedures. We encourage everyone to contribute actively, keep a clear record of their work, and raise concerns about workload distribution early.